

DW03_bictr_decode

Up/Down Binary Counter with Output Decode

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Features and Benefits

- Up/down count control
- Asynchronous reset
- Loadable count register
- Counter enable
- Terminal count flag

Revision History

Description

DW03_bictr_decode is a general-purpose up/down counter whose outputs are binary decoded.

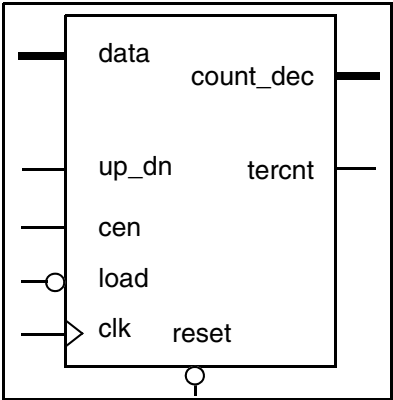


Table 1-1 Pin Description

Pin Name	Width	Direction	Function
data	<i>width</i>	Input	Counter load input
up_dn	1	Input	High for count up and low for count down
load	1	Input	Enable data load to counter, active low
cen	1	Input	Count enable, active high
clk	1	Input	Clock
reset	1	Input	Counter reset, active low
count_dec	2^{width}	Output	Binary decoded count value
tercnt	1	Output	Terminal count flag

Table 1-2 Parameter Description

Parameter	Values	Function
width	≥ 1	Width of data input bus

Table 1-3 Synthesis Implementations

Implementation Name	Function	License Feature Required
str	Synthesis model	DesignWare

Table 1-4 Simulation Models

Model	Function
DW03.DW03_BICTR_DECODE_CFG_SIM	Design unit name for VHDL simulation
dw/dw03/src/DW03_bictr_decode_sim.vhd	VHDL simulation model source code
dw/sim_ver/DW03_bictr_decode.v	Verilog simulation model source code

Table 1-5 Counter Operation Truth Table

reset	load	cen	up_dn	Operation
0	X	X	X	Reset
1	0	X	X	Load
1	1	0	X	Standby
1	1	1	0	Count down
1	1	1	1	Count up

The counter is *width* bits wide, and has 2^{width} states from “000...0” to “111...1”. The counter is clocked on the positive edge of *clk*.

The *reset*, active low, provides for an asynchronous reset of the counter to “000...0”. If the *reset* pin is connected to ‘1’, then the reset logic is not synthesized, resulting in a smaller and faster counter.

The *up_dn* input controls whether the counter counts up (*up_dn* high) or down (*up_dn* low), starting on the next positive edge of *clk*.

The counter is loaded with *data* by asserting *load* (low) and applying data to *data*. The data load operation is synchronous with respect to the positive edge of *clk*.

When the count enable pin, *cen*, is high, the counter is active. When *cen* is low, the counter is disabled, and *count* remains at the same value.

The *count_dec* is an output bus that ranges from $2^{width-1}$ to 0.

The *tercnt* is an output flag that is asserted one state before the count rolls over. When counting up, *tercnt* is high at *count_dec* = “100...000”. When counting down, *tercnt* is high at *count_dec* = “000.....001”.

Timing Diagrams

Figure 1-1 through Figure 1-4 on page 4 show various timing diagrams for DW03_bictr_decode.

Figure 1-1 Functional Operation: Reset, Load, and Count Enable

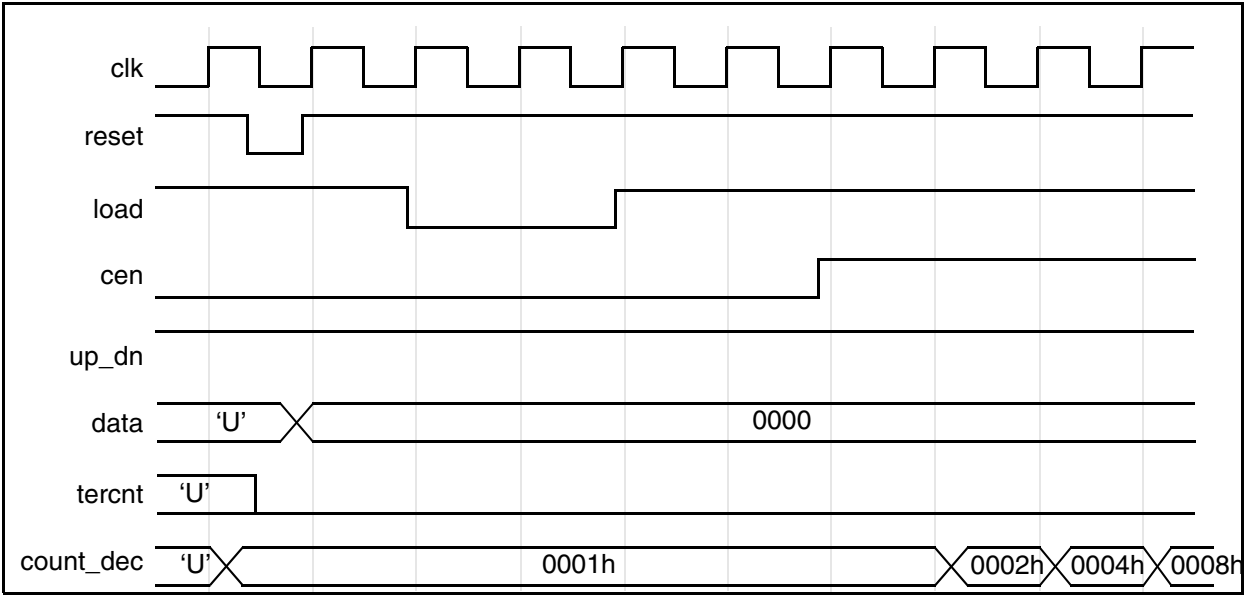


Figure 1-2 Functional Operation: Up and Down Counting

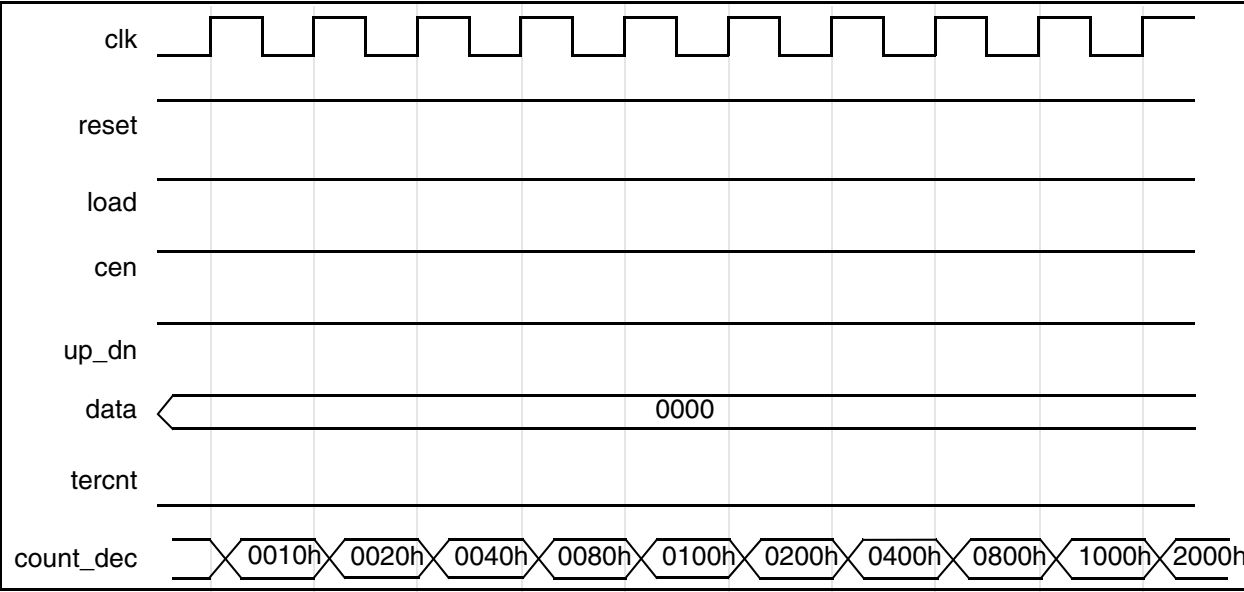


Figure 1-3 Functional Operation: Terminal Count Flag

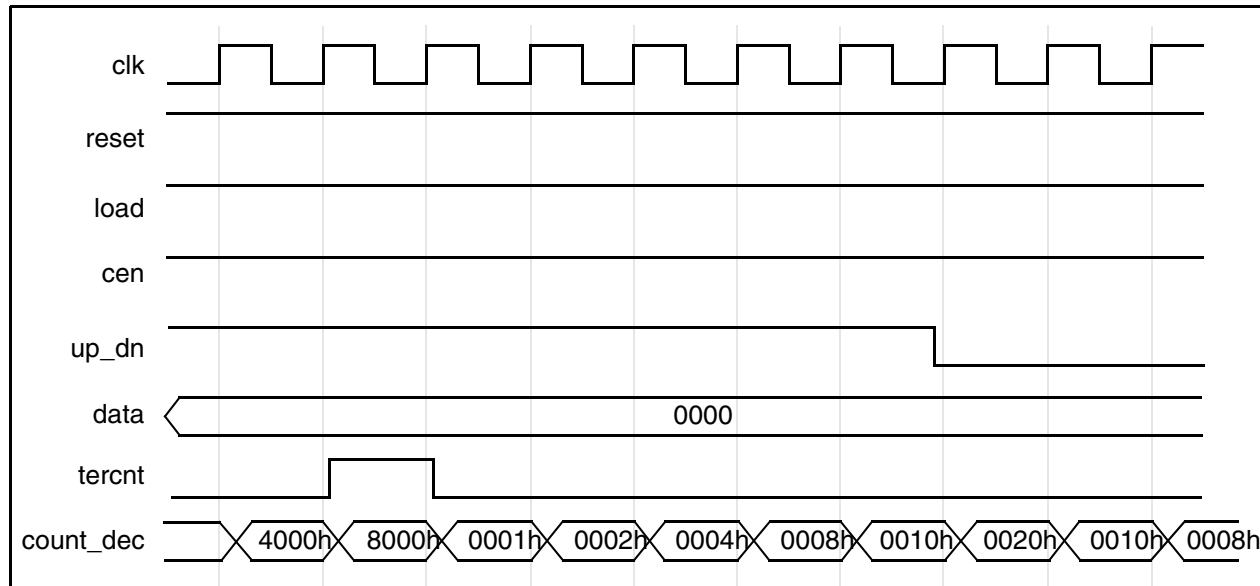
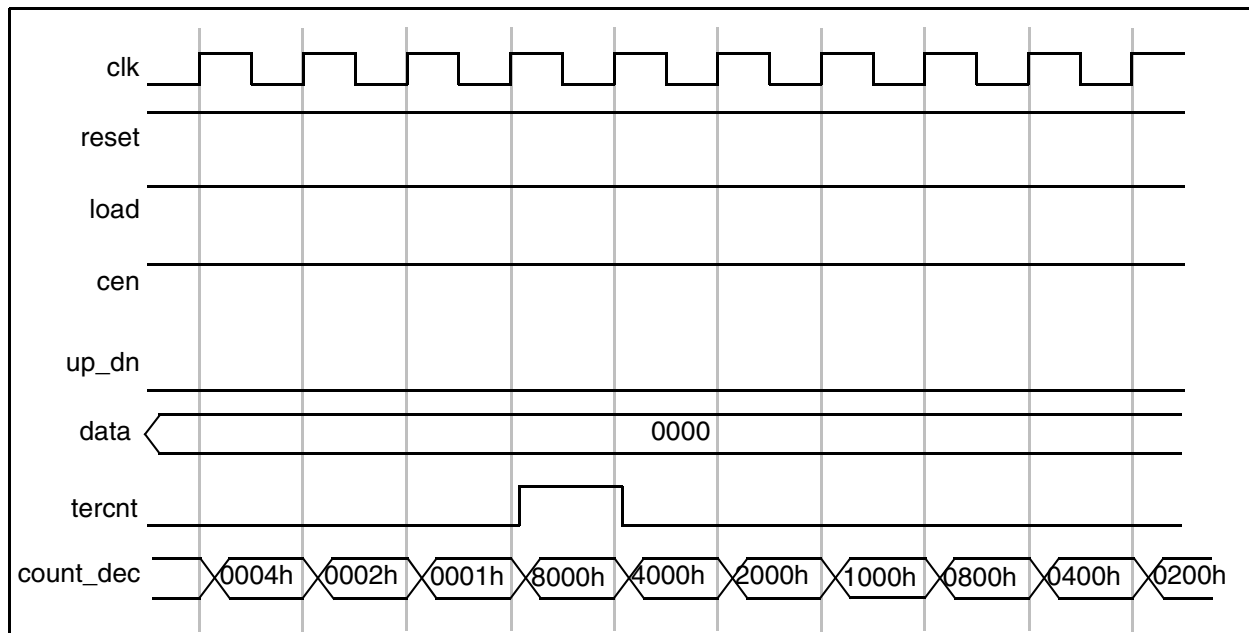


Figure 1-4 Functional Operation: Effect on Terminal Count



Related Topics

- [Logic – Sequential Overview](#)
- [DesignWare Building Block IP User Guide](#)

HDL Usage Through Component Instantiation - VHDL

```
library IEEE,DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DWpackages.all;
use DWARE.DW_foundation_comp.all;

entity DW03_bictr_decode_inst is
  generic (inst_width : POSITIVE := 8);
  port (inst_data      : in std_logic_vector(inst_width-1 downto 0);
        inst_up_dn     : in std_logic;
        inst_load      : in std_logic;
        inst_cen       : in std_logic;
        inst_clk       : in std_logic;
        inst_reset     : in std_logic;
        count_dec_inst : out std_logic_vector(2**inst_width-1 downto 0);
        tercnt_inst    : out std_logic);
end DW03_bictr_decode_inst;

architecture inst of DW03_bictr_decode_inst is
begin

  -- Instance of DW03_bictr_decode
  U1 : DW03_bictr_decode
    generic map ( width => inst_width )
    port map ( data => inst_data, up_dn => inst_up_dn,
              load => inst_load, cen => inst_cen, clk => inst_clk,
              reset => inst_reset, count_dec => count_dec_inst,
              tercnt => tercnt_inst );

end inst;

-- pragma translate_off
configuration DW03_bictr_decode_inst_cfg_inst of DW03_bictr_decode_inst is
  for inst
    end for; -- inst
end DW03_bictr_decode_inst_cfg_inst;
-- pragma translate_on
```

HDL Usage Through Component Instantiation - Verilog

```
module DW03_bictr_decode_inst( inst_data, inst_up_dn,
                              inst_load, inst_cen, inst_clk,
                              inst_reset, count_dec_inst, tercnt_inst );

    parameter width = 8;

    input [width-1 : 0] inst_data;
    input inst_up_dn;
    input inst_load;
    input inst_cen;
    input inst_clk;
    input inst_reset;
    output [(1<<width)-1 : 0] count_dec_inst;
    output tercnt_inst;

    // Instance of DW03_bictr_decode
    DW03_bictr_decode #(width)
        U1 ( .data(inst_data), .up_dn(inst_up_dn), .load(inst_load),
            .cen(inst_cen), .clk(inst_clk), .reset(inst_reset),
            .count_dec(count_dec_inst), .tercnt(tercnt_inst) );

endmodule
```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

For a version of this datasheet with visible change bars, click [here](#).

Date	Release	Updates
September 2024	DWBB_202409.0	■ Updated date and version; no technical changes
December 2023	DWBB_202312.0	■ Updated date and version; no technical changes
March 2019	DWBB_201903.0	■ Removed minPower designation from this datasheet
January 2019	DWBB_201806.5	■ Updated example in “ HDL Usage Through Component Instantiation - VHDL ” on page 5 ■ Added this Revision History table and the document links on this page

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